# Understanding Unix

**NUnix · L3 MIAGE · CM1 · 90 minutes**

ISTIC · University of Rennes  2026–2027

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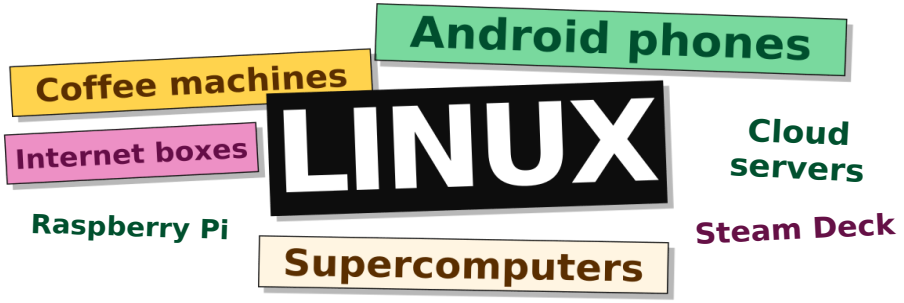
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## Linux is closer than you think

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You can use it all day without opening a terminal.

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## Yes, even the coffee machine

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**Android phones**

The **Linux kernel** manages processes, memory and hardware.

Google · platform architecture



**Freebox Server Pop**

A home internet box. Free publishes its **Linux source code**.

Free · firmware 4.9.10



**MARO Model One**

A coffee machine with **embedded Linux**, remote support and updates.

Toradex · May 2026

Linux can run behind a touchscreen or a web interface.

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## Unix and Linux usage

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**Mobile web use**

**69.17%**

of observed mobile page views come from **Android**, built on Linux.

Statcounter · Sept 2026

**Websites**

**92.1%**

use the **Unix family**, including Linux, among sites whose OS W3Techs identifies.

W3Techs · 1 Oct 2026

**Supercomputers**

**500/500**

systems in the **June 2026 TOP500 list** use the Linux family.

TOP500 · June 2026

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## $ whoami

Who teaches NUnix, and how is the course organised?

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## Louis Ledoux

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**Associate Professor**
University of Rennes · ISTIC
IRISA / Inria · Taran

- Arithmetic circuits: **MLIR (LLVM) / CIRCT**.
- **Systolic arrays** and small floating-point formats.

Previously: Inria / INSA Lyon, Emeraude.
PhD: **BSC / UPC**.
bynaryman.github.io



**MareNostrum 4** · BSC, Barcelona

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## CM, TD and TP

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**NUnix · L3 MIAGE · 2026–2027**

| Sessions | Teaching team |
|---|---|
| **CM** · lectures | Louis Ledoux |
| **TD** · tutorials | Elisabeth Bouriah and Louis Ledoux |
| **TP** · practical sessions | Elisabeth Bouriah and Louis Ledoux |

In TD and TP, we practise the lecture topics through exercises, commands and Bash scripts.

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## Course structure

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**Three lectures (CM) of 90 minutes.**

| Lecture | Topics | Application |
|---|---|---|
| **CM1 · Unix** | Files, paths and permissions | Find files and check access |
| **CM2 · Commands** | Processes, search, pipes and redirections | Count errors in a log |
| **CM3 · Bash** | Variables, conditions, loops and functions | Write scripts with arguments and loops |

**TD and TP:** practise commands first, then write scripts.

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## One line, a useful result

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Which errors occur most often in **1,000 log entries**?

```
sort < errors.log | uniq -c | sort -nr | tee output.txt |
  a2ps -1 -o - | ps2pdf - rapport.pdf
```

Save the counts in `output.txt` and the report in `rapport.pdf`.

**errors.log** · first three lines

```
Database timeout
Login rejected
Connection refused
```

**output.txt** · first three counts

```
    320 Database timeout
    210 Login rejected
    140 Connection refused
```

`a2ps`: text → PostScript
`ps2pdf`: PostScript → PDF

By the end of this course, you will understand this command.

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# 1. Linux and the shell

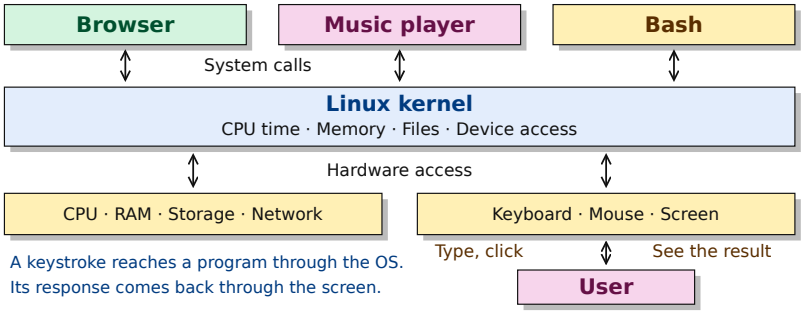
What happens between typing a command and the machine doing the work?

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## Programs, kernel and hardware

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We use hardware to interact with programs.




The kernel **shares resources and controls access**.

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## Linux

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- **1991:** Linus Torvalds announces his kernel project.
- Linux is a **Unix-like kernel**, distributed as free and open source software.
- A distribution combines the kernel with tools, libraries and applications.

Our working environment: a Linux system with the **Bash shell** and command-line tools.

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## Unix, GNU and Linux

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| Name | Meaning |
|---|---|
| **Unix** | The operating-system family that inspired Linux |
| **GNU** | The project behind Bash and many command-line tools |
| **Linux** | The kernel: manages processes, memory and hardware |

**GNU = GNU's Not Unix.**
The name contains itself: a recursive acronym.

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## Touch, click or type

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You already use interfaces to start programs.

| Your action | Interface | Example |
|---|---|---|
| **Touch** a phone screen | Graphical interface | Open an app from its launcher |
| **Click** with a mouse | Graphical interface | Open a folder in a file manager |
| **Type** a command | Text interface | Ask Bash to list a directory |

A smartphone can be a **terminal**: an endpoint for using a service. Its touchscreen, like a mouse or keyboard, provides **input**.

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## Our terminal for this course?

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DEC VT100 · IF-3-SYS, G. Salagnac, INSA Lyon

A **physical terminal** sends keystrokes to a computer and displays its replies. The programs run on the connected computer.

Good news: yours fits in a window.

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## Terminal, shell and command

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| Name | Role |
|---|---|
| **Terminal emulator** | A window for keyboard input and text output |
| **Bash, the shell** | Interprets the line and runs commands |
| **`ls`, a command** | Lists directory entries |

Type `ls` → Bash runs `ls` → the terminal displays its output.

**TTY** = *teletypewriter*: early terminals printed on paper.
Unix keeps the name, even for emulated terminals.

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## A command and its arguments

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```
ls -l /home/alice
```

| Part | Role |
|---|---|
| `ls` | **Command**: list directory entries |
| `-l` | **Option**: change how `ls` lists them |
| `/home/alice` | **Operand**: the directory to list |

Both `-l` and `/home/alice` are **arguments** passed to `ls`.

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## Short and long options

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| Form | Example | Meaning for GNU `ls` |
|---|---|---|
| Short: **one dash** | `ls -a` | Include hidden entries |
| Long: **two dashes** | `ls --all` | Same as `-a` |
| Combined | `ls -la` | `-l` and `-a` together |

Some options take a value: `head -n 5 notes.txt` prints five lines.

A **bare `--` ends options**.
`ls -- -notes.txt` treats `-notes.txt` as a filename.

Each command defines its options. Check `--help` or `man`.

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## Getting help

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```
man ls
info coreutils
apropos directory
help cd
```

- **Tab** completes a name or displays possible completions.
- **Up** recalls a previous command. `history` lists history.
- `clear` clears the display. In `man`, **q** quits.

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# 2. Files and directories

Where is my file, and what can I do with it?

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## One directory tree

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A **directory** is a folder: it contains names of files and other directories.

```
/
├── home/
│   └── alice/
│       └── notes.txt
└── etc/
```

The common root is /. The path `/home/alice/notes.txt` follows the tree.
Alice's **home directory** is `/home/alice`; the university may use a different path.

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## Personal files and system data

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Linux separates personal files, configuration and application data.

| Directory | Typical contents |
|---|---|
| `/home/alice` | Alice’s personal files |
| `/etc` | System configuration |
| `/var` | Changing data: logs, caches, application state |
| `/var/lib` | Persistent application state |
| `/var/log` | Log files |

`/var/lib` and `/var/log` are subdirectories of `/var`.

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## Users, administrators and software

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| Directory | Typical role |
|---|---|
| `/home`, `/root` | User homes; the root account's home |
| `/usr` | Most distribution-provided tools and libraries |
| `/usr/local` | Software installed locally by the administrator |
| `/opt` | Add-on software packages |

`/` is the filesystem root.
`root` is also the administrator account's name.

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## Temporary files and system information

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| Directory | Purpose / lifetime |
|---|---|
| `/var` | Changing data, often persistent |
| `/tmp` | Temporary files; cleanup policy varies |
| `/run` | Runtime state, cleared at boot |
| `/dev` | Device interfaces |
| `/proc`, `/sys` | Views of processes, kernel and devices |

A path can expose live system information.
It need not refer to ordinary data stored on disk.

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## Several filesystems, one tree

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```
/
├── home/       user files
├── mnt/        a temporary mount point
└── media/      removable media mount points
```

Mounting makes another filesystem visible at a directory.
The device node might be `/dev/sdb1`; its files appear at the mount point.

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## The current directory

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| Notation | Meaning |
|---|---|
| `.` | Current working directory |
| `..` | Parent directory |
| `/` | Root directory |
| `~` | Your home directory, expanded by the shell |
| `$HOME` | Variable containing your home path |
| `$PWD` | Variable containing the current directory path |

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## Absolute and relative paths

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Current directory: `/home/alice`

| Path | Interpretation |
|---|---|
| `/home/alice/notes.txt` | Absolute: start at `/` |
| `notes.txt` | Relative: start in the current directory |
| `./notes.txt` | The same relative path |
| `../bob/notes.txt` | Go to the parent, then into `bob` |

A path tells us **where to look**. Permissions determine whether access is allowed.

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## Moving around

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```
pwd
cd /home/alice
cd ..
cd
```

- `pwd` prints the current working directory.
- `cd path` changes the current directory.
- `cd` or `cd ~` returns to your home.

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## Listing directory entries

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```
ls
ls -l
ls -a
ls -la /home/alice
```

- `-l` adds details, including ownership and permissions.
- `-a` includes names beginning with `.`, such as `.bashrc`.
- `ls directory` lists that directory's contents.

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## Creating and removing directories

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```
mkdir practice
rmdir practice
```

`rmdir` removes an **empty** directory.

`rm -r practice` also removes the directory's contents.
Shell deletion does **not** send files to a recycle bin.

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## Creating, copying and moving files

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```
touch notes.txt
cp notes.txt notes-copy.txt
mv notes-copy.txt archive.txt
rm archive.txt
```

`touch` creates a missing file, or updates timestamps of an existing file.

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## Reading text files

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| Command | Result |
|---|---|
| `cat notes.txt` | Print the whole file |
| `less notes.txt` | Browse interactively, **q** to quit |
| `more notes.txt` | Browse a page at a time |
| `head -n 5 notes.txt` | Print the first five lines |
| `tail -n 5 notes.txt` | Print the last five lines |
| `diff old.txt new.txt` | Show differences between two files |

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## Why is less called less?

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Mark Nudelman was investigating errors in large logs.
He wanted to read the lines **before** an error.

His version of `more` could not move backward.
The new pager's name joked about a **backward more**.

```
less app.log
```

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## Concatenating files

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```
cat first.txt second.txt
cat
cat first.txt - second.txt
```

- With filenames, `cat` prints their contents **in argument order**.
- Without filenames, `cat` reads **standard input**.
- A standalone `-` means **read standard input here**.
- At an empty terminal input line, **Ctrl+D** indicates end of input.

The third command reads `first.txt`, then your keyboard input, then `second.txt`.

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## A space can change the arguments

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We want to read **one file** named `meeting notes.txt`.

**Without quotes**

```
cat meeting notes.txt
```

`cat` receives **two filenames**:

- `meeting`
- `notes.txt`

**With quotes**

```
cat "meeting notes.txt"
```

`cat` receives **one filename**:

- `meeting notes.txt`

Quotes keep the filename together. The surrounding `"` characters are **not passed to `cat`**.

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## Three ways to protect a space

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Each command passes the same filename: **`meeting notes.txt`**.

| Form | What you type |
|---|---|
| Single quotes | `cat 'meeting notes.txt'` |
| Double quotes | `cat "meeting notes.txt"` |
| Backslash | `cat meeting\ notes.txt` |

The surrounding quotes and the backslash protect the space. They are **not part of the filename**.

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## Patterns select matching filenames

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The directory contains only **a.txt** and **meeting notes.txt**.

```
cat *.txt
```

The shell finds the matches, then passes **two filenames** to cat: a.txt and meeting notes.txt.

| Pattern | Matches |
|---|---|
| * | Zero or more characters |
| ? | Exactly one character |
| [abc] | One character from the set |

The space **inside a matched filename** stays intact.

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## Quotes keep a pattern literal

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Same directory: **a.txt** and **meeting notes.txt**.

**Let the shell find matches**

```
cat *.txt
```

cat receives **two filenames**:
- a.txt
- meeting notes.txt

It reads both files.

**Protect the star with quotes**

```
cat '*.txt'
```

cat receives **one filename**:
- *.txt

No file has that literal name here: cat reports an error.

cat "*.txt" and cat *.txt also pass a literal *.

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## Single quotes or double quotes?

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`HOME` stores your home directory, here `/home/alice`. `$HOME` asks for its value; `echo` prints its arguments.

| What you type | What you see |
|---|---|
| `echo '$HOME'` | `$HOME` |
| `echo "$HOME"` | `/home/alice` |

**Single quotes** keep `$HOME` literal. **Double quotes** allow it to expand. Both protect spaces and filename pattern characters such as `*`.

With double quotes, we can use the variable and protect the space:

```
cat "$HOME/meeting notes.txt"
```

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## Checkpoint: files and paths

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1. What does `cd ..` do?
2. Which `ls` option includes hidden entries?
3. What does `cat a.txt b.txt` print first?
4. Which filename does `cat '*.txt'` try to open?

1. Move to the parent directory.
2. `-a`
3. The contents of `a.txt`.
4. The literal filename `*.txt`. The shell does not expand the star.

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# 3. Who may access this file?

Why can I read this file, but cannot change or delete it?

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## Users, groups and ownership

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```
whoami
id
ls -ld .
ls -l
```

- `id` shows your user identity and groups.
- `ls -l` shows the file's owner, group and permission bits.
- `ls -ld` inspects a directory itself.

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## Permission classes

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```
-rwxr-xr--
  u  g  o
```

| Class | Applies to |
|---|---|
| u | The file's owner |
| g | Members of the file's group |
| o | Other users |
| a | All three classes when using `chmod` |

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## What r, w and x allow

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| Bit | Regular file | Directory |
|---|---|---|
| **r** | Read contents | List entry names |
| **w** | Modify contents | Add or remove entries* |
| **x** | Execute | Search / traverse |

*Changing directory entries also requires search permission. Other restrictions, such as the sticky bit, may apply.

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## Changing permission bits

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```
chmod u+x script.sh
chmod go-w notes.txt
chmod u=rw,g=r,o= notes.txt
```

- `+` adds the selected permission bits.
- `-` removes them.
- `=` sets the selected class to the listed permissions.

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## Numeric permissions

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Each class adds its bit values: **r = 4**, **w = 2**, **x = 1**.

| Owner | Group | Others |
|---|---|---|
| `rwx` = **7** | `rw-` = **6** | `r--` = **4** |

```
chmod 764 notes.txt
```

The three digits describe **owner, group, others**, in that order.

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## Checkpoint: permissions

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1. Decode `640` for a regular file.
2. Add execute permission for the owner of `run.sh`.
3. Which directory bit allows traversal?

1. Owner: `rw-`
   Group: `r--`
   Others: `---`
2. `chmod u+x run.sh`
3. `x`

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## “I can see the name, but cannot read it”

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```
# Illustrative server metadata
 drwxr----- alice students  shared/
 -rw-r----- alice students  shared/notes.txt
```

Bob belongs to `students`. He can list the names, but cannot open `notes.txt`. Why?

The directory’s group bits lack **x**: traversal.
If team access is intended, the owner can add `g+x`.

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## CM1 summary: files and permissions

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1. Locate configuration, software, data and logs.
2. List and read files; quote names containing spaces.
3. Check owner, group and other permissions.

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## Optional reference

Examples to revisit during TD or personal practice

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## Operating system, shell and GUI

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| Layer | What it gives us |
|---|---|
| **Operating system** | Processes, storage and access control |
| **Graphical interface** | Windows, menus and direct visual interaction |
| **Shell** | Commands we can combine and save in scripts |
| **Application** | A task-specific tool, such as Excel or a database |

Your laptop can run Windows while the company’s application runs on Linux. You need to work with both.

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## Summarise a log before opening it in Excel

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| Quantity | Meaning |
|---|---|
| **2,000,000 log lines** | Example input size |
| **1,048,576 rows** | The maximum in one Excel worksheet |
| **One row per error code** | The summary the team wants to discuss |

Process the raw log where it lives, save the counts, then use Excel to analyse or present the summary.

Microsoft: worksheet limits. Power Query and the Data Model offer other ways to handle larger datasets.

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## Path components and word counts

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```
dirname /home/alice/notes.txt
basename /home/alice/notes.txt
wc -w notes.txt
```

- `dirname` prints `/home/alice`.
- `basename` prints `notes.txt`.
- `wc -w` counts **words**. `wc -l` counts newline characters.

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## Hard links and symbolic links

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### Hard link

```
ln notes.txt other.txt
```

- Another name for the **same file**.
- Removing one name leaves the other usable.
- Same filesystem, normally no directory links.

### Symbolic link

```
ln -s notes.txt shortcut
```

- Stores a **path** to a target.
- May cross filesystems or target a directory.
- Can become dangling if the target path disappears.

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## Builtins and external commands

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### Builtin

Implemented inside the shell.

```
cd ..
```

Can change the current shell's state.

### External program

Loaded from an executable file.

```
ls -l
```

The shell searches directories in PATH.

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## Recognising a command

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```
type cd
type ls
type -a echo
help cd
```

`type` reports how Bash resolves a name.
`help` documents its builtins.

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# Seeing each argument with printf

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Only two files: **`a.txt`** and **`meeting notes.txt`**.

`printf '<%s>\n'` displays each following argument on its own line.

```
printf '<%s>\n' *.txt
```

```
<a.txt>
<meeting notes.txt>
```

**Two filename arguments.**

```
printf '<%s>\n' '*.txt'
```

```
<*.txt>
```

**One literal argument.**

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